

Definition for this bus. The resulting bus interface definition would look like what's shown in Fig. 4.

The next thing to specify is the interfacing mode for the bus interface. The most typical setting is either master or slave mode, and can be found from the Interface mode section. Here the Nios2 processor acts as master on the 'avalon_bus.' So select Master from the pull-down menu shown in 'Error! Reference source not found.' You can add more bus interfaces to it using the same basic principle.

We can identify the created component as a CPU component by adding address space to it. Select Address Spaces from the list given on the left. Add a new address space to the address space summary list, and fill the cells as shown in Fig. 5. Now the component can be identified as a programmable one. Select CPUs, double click the empty area to add a new line to the list, give a name to it, select the address space reference to the created 'avalon_addr_space' and save the component.

Creating channel components

This component is pre-given in the exercise work library.

To make HIBI bus, we need to make a special component that represents the HIBI interconnection network. Create normal components with product hierarchy IP, and fill the VLVN fields as G<group number>, 'ip.hwp.communication,' 'hibi,' and 'draft.' The following bus interfaces need to be added to the component:

1. Slave interface to system_clk_bus
2. Slave interface to rst_n_bus
3. Mirrored master interface to hibi_bus
4. Second mirrored master interface to hibi_bus
5. Third mirrored master interface to hibi_bus

On creating the required bus interfaces, select Channels from the list on the Nios2 tab, and press the '+' button to create a new channel. Enter 'hibi_Channel' in the Name field. Dou-



Fig. 5: Address space addition

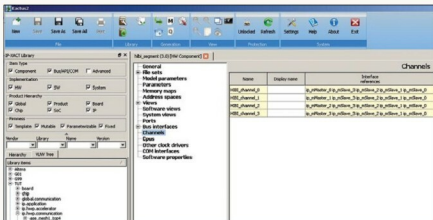


Fig. 6: Creating channel components

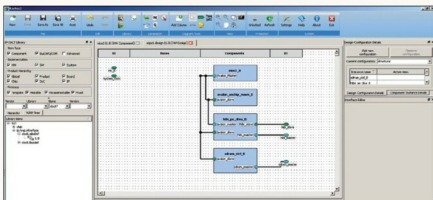


Fig. 7: Creating hierarchical design

ble click on the Bus Interference Reference field and select all hibi_bus bus interfaces to be added to the channel. Now the window will look like what's shown in Fig. 6.

Creating a hierarchical design

When all components and buses are made, we can proceed to the actual hierarchical system design. Go to File→New, select HW design from the pop-up, select SoC from the Product Hierarchy and Mutable from the Firmness pull-down menu. Fill in the VNLV tuple and click on OK to create

a new hardware design.

Drag and drop different components to the design using the Hierarchy or VLVN Tree menu on the left. Tools are provided in the Diagram Tools in the top pane for creating ports and connecting components together with buses.

Some of the basics of Kactus2 have been covered here. By successfully completing the exercise work systems with Kactus2, you can use available IP-XACT components, create your own components, or use draft components. ●

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